

Chapter 7

A reconstruction of Proto-Sawabantu

Rebecca Grollemund^a, Gérard Philippon^b, Braden Brown^a & Chaeli Rule^c

^aUniversity of Missouri ^bLaboratoire Dynamique Du Langage, CNRS ^cThe Ohio State University

The present paper aims at studying and comparing Bantu languages spoken along the Cameroonian, Equatorial Guinean and Gabonese coasts called the Sawabantu languages by reconstructing proto-Sawabantu (consonants and vowels) via the Comparative Method used in Historical Linguistics. The proto-Sawabantu reconstructions obtained are analyzed and compared with other proto-languages reconstructed for languages spoken in this region: proto-Bantu and proto-Manenguba in order to identify Sawabantu lexical innovations and phonological innovations. The resulting historical analyses shed new light on the fascinating history of the speakers of this particular region where the great Bantu Expansion started 5,000 years ago in southern Cameroon.

1 Introduction

Sawabantu Bantu languages are located along the Cameroonian, Equatorial Guinean and Gabonese coasts (Figure 1) and generally include Guthrie's (Guthrie 1967–1971, later updated by Maho n.d.) A10, A20 and A30 languages. Cameroon is well-known for its significant cultural and linguistic diversity with approximately 250 spoken languages (Eberhard et al. 2019) belonging to three different language families: Niger-Congo (190 languages including the Bantoid and Bantu languages), Nilo-Saharan (2 languages) and Afro-Asiatic (55 languages). Moreover, it is in southern Cameroon that the Bantu Expansion started 5,000 years ago with Bantu-speaking-populations migrating southward through Cameroon, Equatorial Guinea and Gabon expanding later across Central, Eastern and Southern Africa (Bostoen 2007, Grollemund et al. 2015, Koile et al. 2022).



Several studies have been conducted on Sawabantu languages highlighting their relatedness, with A10-20-30 languages (minus Manenguba A15 and Bubi A31) forming a valid genetic group. Ebobissé (1989) proposed first a dialectometrical study of A20 and A30 Sawabantu languages for which he calculated the percentages of linguistic proximity based on the study of lexicon. His results divided the Sawabantu clade into two subgroups with A20 languages on one hand and A30 languages on the other hand. In 2015, Ebobissé (2015) furthered his study of Sawabantu languages by providing a detailed synchronic study of the same languages with a focus on phonology, morphology and lexicon confirming the two subgroups identified in 1989. The addition of A10 languages (minus Manenguba A15) sometimes called “Oroko” (Friesen 2002) to the Sawabantu clade was made by Nurse & Philippson (2003: 170) who established a classification of 80 Bantu languages based on the study of non-lexical data (30 phonological and morphological features). The study of lexicon but also morphological criteria such as the shape of the nominal stem that is generally CVCV in A10-20-30 as opposed to Manenguba A15 (Philippson 2022: 6-7) were decisive in their classification. Based on these findings, Philippson (2022) proposed to borrow the geographical term “Sawabantu” from Ebobissé (2015) to refer to these A10-20-30 languages that have a genetic value.

The purpose of this paper is to examine Sawabantu languages and reconstruct their putative ancestor Proto-Sawabantu (PSB) by the application of the Comparative Method (CM). PSB will be analyzed and compared with other proto-languages reconstructed in this region such as Proto-Manenguba (PM) reconstructed by Hedinger (1987) that includes A15 languages spoken near the Bantu cradle. The comparison between PSB and PM reconstructions reveals two different evolutionary trajectories with PSB being more similar to Proto-Bantu (PB) in certain aspects as opposed to PM confirming therefore the differences observed between Manenguba A15 and Sawabantu A10-A20-A30 languages. PSB reconstructions will also be compared to PB reconstructions taken from BLR3 (Bastin et al. 2002) in order to identify any diachronic phonological innovations that will shed new light on the peculiar evolution of the languages spoken in this region.

2 Data and methods

2.1 Languages

Based on the study of lexicon (Ebobissé 1989, 2015) and non-lexical data such as phonological or morphological features (Nurse & Philippson 2003: 70), the Sawabantu clade generally comprises between 20–27 Bantu languages belonging

to Guthrie's zones A10, A20 and A30 (e.g., Table 1) with the exception of the A15 languages called Manenguba and A31 Bubi spoken on Bioko Island.

In order to start our comparative work, we selected the same A20 and A30 Sawabantu languages provided in Ebobissé (2015) study. Ebobissé (2015) divided the Sawabantu languages into three groups based on geographical locations: South (Banoho, Batanga, Yasa, Bapuku), Central (Duala, Bidiman, Oli, Pongo, Mongo, Mulimba) and Southwest (Mboke, Mokpe, Isu, Kole, Bubia) but his classification based on the study of lexicon published in 1989 showed that Central and Southwest were more closely related. We will use in this paper the same geographical distinctions established in Ebobissé (1989) and Ebobissé (2015) with the mention of South, Central, Southwest with the addition of the North region ("Oroko") represented in our study by the Kundu language (A10) that we have extracted from Friesen (2002) in order to get a more representative sample of the Sawabantu languages.

In his monumental study, Ebobissé (2015) collected, compared and analyzed 610 words for 15 languages. For the purpose of this study, we created an Excel database containing all 610 words for the same 15 languages with the addition of Kundu, an A10 language. Words from Ebobissé were translated from German to English. In addition to collecting the individual words, we searched for corresponding PB words drawn from BLR 3 (Bastin et al. 2002) for comparison purposes.

2.2 Lexicon-based classifications of Sawabantu languages

Bantu languages spoken in southern Cameroon belong to the North-Western (NW) genetic group (Bastin et al. 1999, Grollemund 2012, Currie et al. 2013, Grollemund et al. 2015, Koile et al. 2022) that usually includes all the A languages (Guthrie's (1967–1971) referential zones) generally with B10-20-30 languages spoken in Gabon. Bastin et al. (1999), for example, include in their NW group A languages (without Mbam-Bubi composed of A31-some A40-50-60 languages) and B10-20-30 languages. According to Bastin et al. 1999's classification, Mbam-Bubi languages are the first group to diverge followed by the NW. Grollemund et al. (2015) shows that NW was composed of several clades that generally includes: Mbam-Bubi (+Jarawan), A10-20-30-40+A70-80-90 (labeled "NW Cameroon"), some B20+A80-90, remaining B20 and B10-30 (labeled "NW Gabon"). The latest phylogenetic classification established by Koile et al. (2022) proposes a different result wherein NW is composed of a sole node that includes only A languages and some of B20; whereas the remaining B20 and B10-30 languages are classified

Table 1: Sawabantu languages in Ebobissé (2015) with selected languages for this study in italics with Maho’s codes (n.d.) and ISO codes when available (some languages do not have ISO code because they are either understudied/unknown or considered as some variants of another language).

Region	Language	Maho (2009)	ISO 639-3
South	<i>Banoho (BAN)</i>	A32a	
	<i>Batanga (BAT)</i>	A32	bnm
	<i>Yasa (YAS)</i>	A33a	yko
	<i>Bapuku (BAP)</i>	A32b	bng
	Kombe	A33b	nui
	Benga	A34	bng
Central	<i>Duala (DUA)</i>	A24	dua
	<i>Bidiman (BID)</i>	A241	
	<i>Oli (OLI)</i>	A25	
	<i>Pongo (PON)</i>	A26	
	<i>Mongo (MON)</i>	A261	
	<i>Mulimba (MUL)</i>	A27	mzd
Southwest	<i>Mboko (MBO)</i>	A21	mdu
	<i>Mokpe (KPE)</i>	A22	bri
	<i>Isu (ISU)</i>	A23	srz
	<i>Kole (KOL)</i>	A231	kme
	<i>Bubia (BUB)</i>	A221	bbx
North	Londo	A11	
	Ngolo	A111	
	Bima	A112	
	Batanga	A113	
	Bakoko	A114	
	Londo	A115	
	Lue	A12	
	Mbonge	A121	
	<i>Kundu</i>	A122	
	Ekombe	A123	

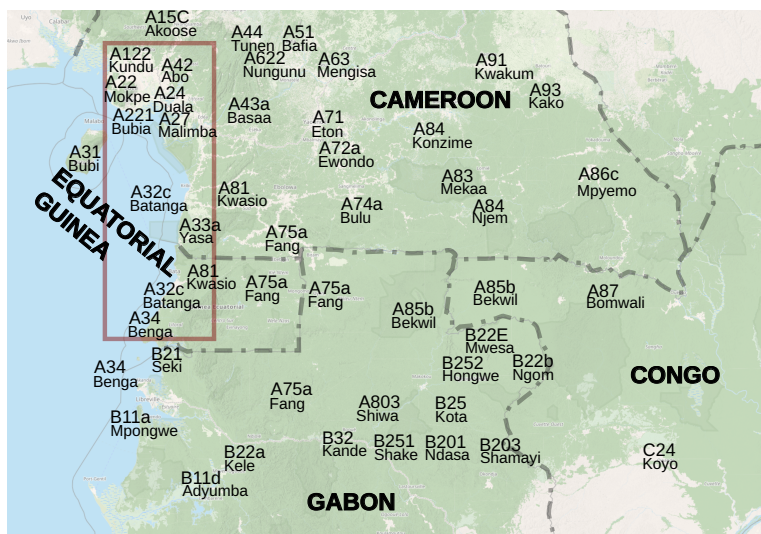


Figure 1: Approximate location of Sawabantu languages (in the red rectangle)

into a different clade. To sum up, Sawabantu languages belong to the first group of Bantu languages to diverge after the Mbam-Bubi group.

A closer look at the Bantu trees with a focus on the NW clades (Figure 2) established by Grollemund et al. (2015) and Koile et al. (2022) confirms that Mbam languages (A40-50-60) are the first Bantu languages to diverge followed by a group that includes A10-20-30-40-70 (in Grollemund et al. 2015) or only A10-20-30 (in Koile et al. 2022). In both trees, Sawabantu languages split after the Mbam languages and form a clade with Manenguba (A15) languages confirming their relatedness. In addition, both classifications highlight a noticeable division amongst Sawabantu languages with a first group composed of A24-A27-A33a-A34 (corresponding to Ebobissé's South and Central groups) and a second group composed of A22-A122 languages (Southwest) linked with two A10 languages Londo and Mbonge (North). This observation contradicts Ebobissé's (1989) results where Central and Southwest are more closely related but confirms Nurse and Philippson's (2003) classification with the inclusion of North Sawabantu languages within the Sawabantu group. Finally, we observe that this last subgroup (A22-A122-A11-A121) is linked to the Manenguba languages (A15) as a coordinated branch.

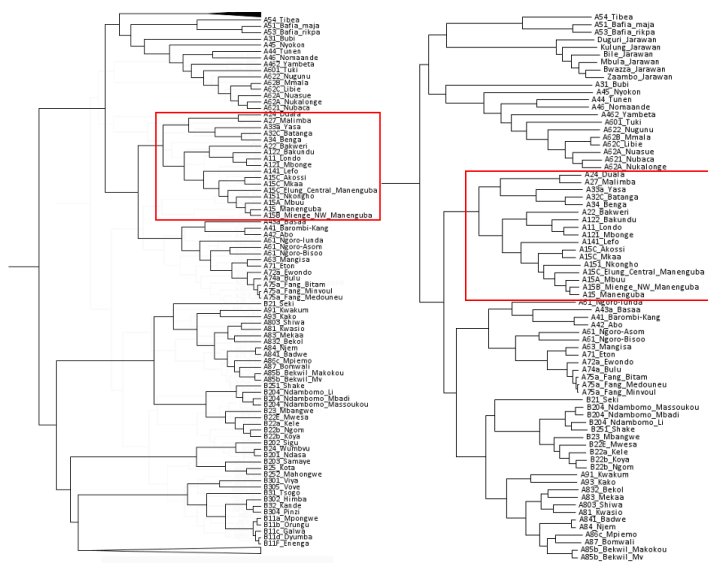


Figure 2: Comparison of the two NW groups in Grollemund et al. (2015) and Koile et al. (2022)

2.3 Method

In order to reconstruct PSB, we applied the Comparative Method (CM) and its series of steps. This upstream approach utilizes the systematic comparison of cognate words in related languages to reconstruct the linguistic properties of the putative ancestor with a great degree of certainty (Campbell 1998). In order to successfully apply the CM, several steps must be completed. First, the identification of cognates - words that are similar in form and meaning that are assumed to have been inherited from a common ancestral source - is paramount. We studied each meaning to identify cognate sets¹. Most of the words studied yielded between one to three cognate sets (e.g., Table 2 ‘two’, ‘house’ and ‘surpass’) whereas, in rare cases, meanings would yield as many as five different cognate sets (e.g., Table 2 ‘yesterday’).

Once all the cognate sets were identified, we proceeded to the next step which consisted of the alignment of all the cognate Bantu roots in C1, V1, C2 and V2 positions in order to list all the sound correspondences present in the data. The sound

¹Because of the space restriction, we are not able to present our data with our 600 comparative series. However, many examples are presented in this paper with approximately 100 different comparative series for the languages studied.

Table 2: Examples of cognacy judgement, numbers refer to cognate sets. The capital letters refer to the geographical denominations with “S” for South, “C” for Central, “SW” for Southwest and “N” for North. Empty cells mark missing data.

Meanings	two	house	breast	surpass	yesterday
BAN - S	bé-fà 1	ndáfbò 1	di-bée 1	i-bála 1	váké 1
BAT - S	bé-fà 1	ndáfbò 1	di-bée 1	i-bála 1	váhé 1
YAS - S	bí-fà 1		i-béne 1	e-bále 1	
BAP - S	bé-fà 1	ndáfbò 1	i-bée 1	i-bála 1	
DUA - C	bé-fàá 1	ndáfbò 1	di-bée 1	búka 2	
BID - C	bí-fàá 1	ndáfbò 1	bée 1	búa 2	fwéná 2
OLI - C	bí-fàá 1	ndáfbò 1	bée 1	búa 2	fwéá 2
PON - C	bé-fàá 1	ndáfbò 1	i-bée 1	i-búka 2	mbánā 2
MON - C	bé-fàá 1	ndáfbò 1	di-bée 1	i-búka 2	kéke 3
MUL - C	bé-fàá 1	ndáfbò 1	di-bée 1		céce 3
MBO - SW	bé-βàá 1	ndáwò 1	li-βée 1	laáka 3	
KPE - SW	bé-βàá 1	ndáwò 1	li-βée 1	laáka 3	ημέλεε 4
ISU - SW	bé-fàá 1	ndáwò 1	li-βée 1	yaáka 3	kéekée 5
KOL - SW	bé-bàá 1	ndáfbò 1	di-bée 1	yaáka 3	kéekée 5
BUB - SW	bé-βàá 1	ndáwò 1	li-βée 1	láka 3	ημέλεε 4
Kundu - N	βε-βε 1	ndáβo 1	di-bê 1		jana 2

correspondences (study of the phonological similarities between the languages studied) were then organized to facilitate the identification of regular sound correspondences. If a sound correspondence occurs several times, it suggests that the sound correspondence is regular which implies that the sounds could have either been inherited from proto-sounds located at an ancestral level or innovated. For example, if in all the Sawabantu languages we found the sound /m/ in C1 and C2 positions, it is reasonable to assume that the PSB had *m in the same positions. The establishment of regular sound correspondences is then used to reconstruct ancestral sounds. Finally, once the regular sound changes have been understood via the study of regular sound correspondences, historical linguists can proceed with the reconstruction of words.

3 Results

3.1 Comparative tables

In this section, we present the consonant and vowel sound correspondences as well as the reconstructed sounds labeled “PSB”. The reconstructions made are primarily based on noun and verb stems which generally present a CVCV structure for nouns and CVC structure for verbs in Sawabantu languages.

3.1.1 Vowels in C1 and C2 positions

We reconstructed a system with seven vowels at the PSB level. Examining regular sound correspondences for vowels (presented in Table 3) shows that vowels are highly regular in Sawabantu languages. This regularity contrasts with Manenguba languages where vowels have undergone a number of changes since PM such as assimilation, raising or lowering which complicate the reconstruction according to Hedinger (1987: 44-45). Hedinger (1987: 45-46) also reconstructs long vowels for PM that are generally the result of two distinct processes: (i) loss of the PB *C2 in Manenguba languages has created CVV roots (PB *bèdú ‘colanut’

Table 3: Reconstruction of PSB Vowels

PSB	*i	*e	*ɛ	*a	*ɔ	*o	*u
BAN - S	i	e	ɛ	a	ɔ	o	u
BAT - S	i	e	ɛ	a	ɔ	o	u
YAS - S	i	e	ɛ	a	ɔ	o	u
BAP - S	i	e	ɛ	a	ɔ	o	u
DUA - C	i	e	ɛ	a	ɔ	o	u
BID - C	i	e	ɛ	a	ɔ	o	u
OLI - C	i	e	ɛ	a	ɔ	o	u
PON - C	i	e	ɛ	a	ɔ	o	u
MON - C	i	e	ɛ	a	ɔ	o	u
MUL - C	i	e	ɛ	a	ɔ	o	u
MBO - SW	i	e	ɛ	a	ɔ	o	u
KPE - SW	i	e	ɛ	a	ɔ	o	u
ISU - SW	i	e	ɛ	a	ɔ	o	u
KOL - SW	i	e	ɛ	a	ɔ	o	u
BUB - SW	i	e	ɛ	a	ɔ	o	u
Kundu - N	i	e	ɛ	a	ɔ	o	u

> PM *bèé) and (ii) compensatory lengthening in CVC roots after C2 was deleted (PM *fɪɲ ‘pus’ > /-híi/ in Myenge).

The situation is less complicated in Sawabantu languages where we mostly have CVCV roots for noun stems. There are some languages where non-contrastive long vowels are attested. Generally, they are also the results of the loss of C2 (such as PSB *k) leading to the emergence of CVV roots. In such cases, long vowels or VV sequences can emerge such as in PSB *toko ‘spoon’ > /too/ or PSB *sako ‘army ant’ > /sao/ in some Central Sawabantu languages (where PSB *k2 > Ø). Therefore, because this phenomenon is caused by the deletion of a C2 that is still attested in the remaining Sawabantu languages (as opposed to PM where the PB *C2 has completely disappeared in all Manenguba daughter languages), we chose to not reconstruct long vowels.

The system reconstructed for PSB is a typical PB 7-vowel system with *ɪ and *ʊ for Bantu languages spoken in Central Africa (Guthrie 1967–1971, Meeussen 1967, Bastin et al. 2002). The sole disagreement that existed with the reconstruction of PB vowels was regarding the height of the high vowels *i, *ɪ, *u, *ʊ (Bastin et al. 2002) vs *ĩ, *ĩ, *ũ, *ũ (Guthrie 1967–1971, Meeussen 1967). While Guthrie (1967–1971) and Meeussen (1967) propose that some PB vowels were super close (marked with a cedilla), Bastin et al. (2002) choose to have a contrast between

Table 4: Examples for the reflexes of PSB vowels in V1 and V2 positions. Empty cells represent either missing data or non-cognate words.

PSB meaning	*i ₁₂ cloud	*e ₁₂ tree	*ɛ ₁₂ stand	*a ₁₂ divide	*ɔ ₁₂ knee	*o ₁₂ skin	*u ₁₂ turtle
BAN - S	dĩ-vĩndĩ	yelé	i-témé	i-kaɓa	dĩ-bóŋgɔ	ɲ-hoɓo	kúɖù
BAT - S	dĩ-vĩndĩ	yelé	i-témé	i-kaɓa	dĩ-bóŋgɔ	ɲ-hoɓo	kúɖù
YAS - S	i-vĩndó	e-yilé	e-témé	e-kaɓa	ì-bóŋgɔ	mo-koɓo	kúɖù
BAP - S	dĩ-vĩndĩ	yelé	i-témé	i-kaɓa	ì-bóŋgɔ	ɲ-koɓo	kúɖù
DUA - C	di-wĩndĩ	ɓw-elé	témé	j-aɓa	dĩ-bóŋgɔ	e-yoɓo	wúɖù
BID - C	findifindi	ɓw-elé	témé	y-aɓa	ɗ-òŋgò	o-ɓo	wúlù
OLI - C	vĩndivĩndĩ	ɓw-elé	témé	y-aɓa	ɓóŋ	o-ɓo	wúlù
PON - C	i-wĩndiwĩndĩ	ɓw-elé	i-témé	ij-aɓa	ì-bóŋgɔ	e-yoɓo	wúɖù
MON - C	di-wĩndiwĩndĩ	ɓw-elé	i-témé	-kaɓane	dĩ-bóŋgɔ	e-koɓo	kúɖù
MUL - C	di-ĩndĩ	ɓw-elé	i-témé	i-kaɓa	dĩ-bóŋgɔ	e-yoɓo	kúɖù
MBO - SW	li-wĩndó	eelé	li-témé	li-ɓaá	li-wóŋgóŋgɔ	e-kowo	è-kúlè
KPE - SW	li-wĩndĩ	gb-eé	li-témé	li-ɓaá	li-wóŋgóŋgɔ	yowo	kúlù
ISU - SW	li-wĩndiwĩndĩ	gb-elé	i-témé	i-kaɓa	li-wóŋgóŋgɔ	e-kowo	è-kúlèkúlè
KOL - SW	di-wĩndĩ	gb-eré	i-témé		dĩ-bóŋgɔ		è-kúrè
BUB - SW	li-ĩndó	eelé	li-témé	li-kaɓa	li-wóŋgóŋgɔ	m-owo	è-kúlèkúlè
Kundu - N		bo-le		-kaɓa	dĩ-bóŋgɔ	e-kobo	ku

tense and lax high PB vowels. Some modern-day Bantu languages present the tense/lax opposition ([ɪ] and [ʊ]) whereas other languages, such as Sawabantu languages, have mid vowels ([e] and [o]) instead.

3.1.2 Consonants

We present in this section the consonants reconstructed for PSB. When needed, a distinction between C1 (initial consonant) and C2 (intervocalic consonant) is made (e.g., *k₁ vs *k₂ in Table 5). First, we present the voiceless plosives in Table 5.

Table 5: Sound correspondences for unvoiced plosives. When two reflexes are presented, the first one is the most frequent. Bold characters mark sound changes from PSB to the daughter languages.

PSB	*p ₁₂	*t ₁₂	*k ₁	*k ₁ / _Vk ₂	*k ₁ / _e	*k ₁ / _wV	*k ₂
BAN - S	p	t	h~k	k	k~h	(w)	h~k
BAT - S	p	t	h~k	k	k~h	(w)	h~k
YAS - S	p	t	k	k	c	(w)	k
BAP - S	p	t	k	k	s	(w)	k
DUA - C	p	t	Ø~k	k	Ø	k(w)	Ø~k
BID - C	p	t / r (*t ₂)	Ø~k	k	Ø	k(w)	y~Ø
OLI - C	p	t / r (*t ₂)	Ø~k	k	Ø	k(w)	y~Ø
PON - C	p	t	Ø~k	k	Ø	k(w)	Ø~k
MON - C	p	t	k	k	k	k(w)	k
MUL - C	p	t	Ø~k	k	c	k(w)	Ø~k
MBO - SW	ɸ	t	k	k	k	kp	k
KPE - SW	ɸ	t	Ø~k	k	Ø~k	kp	Ø~k
ISU - SW	ɸ	t	k	k	k	kp	k
KOL - SW	ɸ	t	Ø~k	k	Ø~k	kp	Ø~k
BUB - SW	ɸ	t	Ø~k	k	Ø~k	kp	Ø~k
Kundu - N	ɸ	t	k	k	k	kp	k

For the voiceless plosives presented in Table 5 (examples are presented in Table 6), we reconstruct *p in C1 and C2 positions for the set p/ɸ with South and Central Sawabantu languages displaying /p/ whereas Southwest and North Sawabantu languages have the fricative /ɸ/. Based on the principles of majority wins used in CM and directionality (lenition is more common than fortition, (Bybee & Easterday 2019), *p is the logical choice.

We chose to reconstruct *t in C1 and C2 positions for the set t/r (Oli and Bidiman) because all the Sawabantu languages display /t/ in C1 and C2 positions

Table 6: Examples for the reflexes of PSB *p in C1 and C2 positions. We can notice some voicing occurring in Bidiman with *p₂ > /b/ in the examples presented here (“lip”, “touch”), this phenomenon is irregular as we can find *p₂ > /p/ in between vowels (“sweep”: [papa] in Bidiman).

PSB meaning	*p ₁ viper	*p ₁ chalk	*p ₁ fruit	*p ₁ bury	*p ₁₂ lip	*p ₂ touch	*p ₂ armpit	*p ₂ wind
BAN - S	pée	pémbé	e-ɸumá	i-puɸe	è-pòpò		jájápi	mpupé
BAT - S	pée	pémbé	e-pumá	i-puɸe	è-pòpò	i-tápa	jájápi	mpupé
YAS - S	péyi	pémbé	e-ɸumá				ɲóɲópi	mpupé
BAP - S	pée	pémbé	e-ɸumá	i-puɸe	è-pòpò		náɸè	mpupé
DUA - C	pé	pémbé	e-pumá	pule	è-pòpò	tápa	è-yápi	
BID - C	pé	pémbé	pumá		pòɸò	táɸa		
OLI - C	pé	pémbé	pumá		pòpò	tápa		
PON - C	pé	pémbé	e-pumá	i-pule	è-pòpò	i-tápa	è-yápi	
MON - C	pé	pémbé	e-pumá	i-pule	è-pòpò	i-tápa	è-yápi	
MUL - C	pé	pémbé	e-pumá	i-puɸe	è-pòpò	i-tápa	yàápi	mu-pupé
MBO - SW	ɸée	ɸémbé	e-ɸumá				náɸè	
KPE - SW	ɸée	ɸémbé	e-ɸumá				náɸàɸè	
ISU - SW	ɸé	ɸémbé	e-ɸumá	i-ɸule	è-ɸòɸò	i-táɸà	náɸàɸè	
KOL - SW	ɸée	ɸémbé	e-ɸumá	i-ɸure	è-ɸòɸò	i-táɸà	náɸàɸi	
BUB - SW	ɸée	ɸémbé	e-ɸumá				náɸàɸè	
Kundu - N			e-ɸuma	ɸura				di-ɸeɸe

except for Bidiman and Oli where *t₂ undergoes lenition (rhotacism) only in C2 position (e.g., Table 7 for examples).

For the overlapping sets where we have k/c/s/Ø in C1 and C2 positions, we reconstructed one proto-sound *k based on several factors. First, some comparative sets can be explained by phonological contexts that triggered conditioned sound changes. For example, in C1 position, *k is maintained in Sawabantu languages when the stem presents another velar plosive (*k or *g) in C2 position (e.g., Table 8, examples ‘to be sufficient/to grow’ or ‘molar’) or when the words originally belonged to the PB class 9 (Hyman 2019) with the nasal prefix (e.g., Table 8 ‘pangolin’, reconstructed *kákà in PB, BLR 3 1684, classes 9/10). When followed by a front vowel, *k₁₂ is palatalized in Yasa, Bapuku, Mulimba while the remaining languages present the regular reflexes for *k₁₂ (e.g., Table 8 ‘molar’ and Table 9 ‘muteness’). Finally, we observe some sets for *k₁ with /h/ ~ /k/ in Banoho and Batanga, /k/ in Yasa, Bapuku, Mongo, Mboko, Isu and Kundu, /Ø/ ~ /k/ in Duala, Bidiman, Oli, Pongo, Mulimba, Kole, Mokpe and Bubia (e.g., Table 8, examples ‘divide’, ‘storm’, ‘spear’). The examination of the examples displaying

Table 7: Examples for the reflexes of PSB *t in C1 and C2 positions. Only one example displays *t₁ > /r/ in Oli and Bidiman (e.g., “ear”).

PSB meaning	*t ₁ five	*t ₁ ear	*t ₁ drill	*t ₁ send	*t ₁₂ smoke	*t ₂ bag	*t ₂ hide
BAN - S	bé-táno	ì-tòì	i-túba	i-tila	í-tùtù	ṛ-kùtà	i-huta
BAT - S	bé-táno	ì-tòì	i-túba	i-tila	í-tùtù	ṛ-kùtà	i-huta
YAS - S	ḃi-táno	dì-lóò	e-túba	e-tila	ví-tùtù	mò-kùtà	
BAP - S	bé-táno	dì-tó	i-túba	i-tila	í-tùtù		i-kuta
DUA - C	bé-tánu	tòì	túba	tila	í-tùtù	mù-kùtà	wuta
BID - C	bì-táan	dì-ró	túba	tila	mí-tùrù	ṛ-kùrà	wura
OLI - C	bì-táan	dì-ró	túba	tila	mí-tùrù	ṛ-kùrà	wura
PON - C	bé-tánu	ì-tòì	i-túba	i-tila	í-tùtù	mù-kùtà	i-wuta
MON - C	bé-tánu	ì-tòì	i-túba	i-tila	í-tùtù	mù-kùtà	i-kuta
MUL - C	bé-táan	ì-tòì	i-túba	i-tila	í-tùtù	mù-kùtà	i-kuta
MBO - SW	ḃé-táa	li-tóò	li-túḃa	li-tila	í-tùtù	mò-kùtà	
KPE - SW	ḃé-táa	li-tóò	li-túḃa	li-tilá	yú-tùtù	mò-kùtà	l-utá
ISU - SW	ḃé-tánu	ì-tòì	i-túḃa	i-tila	yú-tùtù	mò-kùtà	i-kuta
KOL - SW	bé-tánu	ì-tòì	i-túba	i-tira	í-tùtù	mò-kùtà	i-wuta
BUB - SW	ḃé-táno	li-tóò		li-tila	í-tùtù	mò-kùtà	l-uta
Kundu - N	be-ta	di-toi		tira	mo-tutu		

Table 8: Examples for the reflexes of PSB *k in C1 position

PSB meaning	*k ₁ divide	*k ₁ storm	*k ₁ spear	*k ₁ / *N_ tree trunk	*k ₁ / _*Vk ₂ /g ₂ sufficient/grow	*k ₁ / *N_ pangolin	*k ₁ / _*ek ₂ /g ₂ molar
BAN - S	i-kaḃa	ḃo-huḃi	di-hoṅgó	ṛ-kòhò		kángà	dì-kehò
BAT - S	i-kaḃa	ḃo-huḃi	di-hoṅgó	ṛ-kòhò	i-kowa	kángà	dì-kehò
YAS - S	e-kaḃa	ḃo-kuḃi	i-kòṅgó	mo-kòkò	e-koka	kángà	i-còkó
BAP - S	i-kaḃa	ḃo-kuḃi	i-kòṅgó	ṛ-kòkò		kángà	i-sekò
DUA - C	j-aḃa	m-udi	j-oṅgó	mu-kòkò	koka	káàn	e-kikò
BID - C	y-aḃa	m-uli	j-oṅgó	ṛ-kòyò	koya	káàn	kiyi
OLI - C	y-aḃa	m-uli	j-oṅgó	ṛ-kòhò	koya	káàn	kiyi
PON - C	ij-aḃa	m-udi	j-oṅgó	mu-kòkò	i-koka	káàn	e-kikò
MON - C	-kaḃaṇ	mu-kudi	j-oṅgó	mu-kòkò	i-koka	káàn	e-kikò
MUL - C	i-kaḃa	m-udi	j-oṅgó	mu-kò		káàn	e-cikò
MBO - SW			di-kòṅgó	mu-kòkò	li-koka	kángè	e-kekò
KPE - SW		mu-uli	li-kòṅgó	mo-kòkò	li-koka	kángè	ε-kòkò
ISU - SW	i-kaḃa	mu-kuli	di-kòṅgó	mo-kòkò	i-koka	kángè	e-kikò
KOL - SW			j-oṅgó	mo-kòkò	i-koka	káàn	e-kikò
BUB - SW	li-kaḃa		li-kòṅgó	mo-kòkò	li-koka	kángè	ε-kòkò
Kundu - N	-kaḃa		di-kòṅgò	ε-kòkò (A11)			e-kekò

Table 9: Examples for the reflexes of PSB *k in C2 position

PSB meaning	*k ₂ elephant	*k ₂ one	*k ₂ scoop	*k ₂ fog	*k ₂ / _I muteness	*k ₂ cow	*k ₂ mortar	*k ₂ spoon
BAN - S	n-jɔhu	y-ɔhɔ	i-tóka	mbàhà	dí-mbúhé	ɲaka	è-bòwà	tóhò
BAT - S	n-jɔhu	y-ɔhɔ	i-tóka	mbàhà	dí-mbúhé	ɲaka	è-bòwà	tóhò
YAS - S	sɔku	e-vɔkɔ		mbàkà	imbúcé	ɲaka	è-bòkà	tókò
BAP - S	n-jɔku	y-ɔkɔ	i-tóka	mbàkà	imbúsé	ɲaka	è-bòkà	tókò
DUA - C	n-jɔu	e-wɔ	tóa	mbà	mbúké	ɲaka	è-bòkí	tòò
BID - C	n-jɔ	fɔ	tóa	mbà	mbúyí	ɲaya	bòyí	tòò
OLI - C	n-jɔ	wɔ	tóa	mbà	mbúyí	ɲaya	bòyí	tòò
PON - C	n-jɔu	e-wɔ	i-tóa	mbà	mbúké	ɲaka	è-bòkí	tòò
MON - C	n-jɔku	y-ɔkɔ	i-tóka	mbàkà	mbúké	ɲaka	è-bòkí	tókò
MUL - C	n-jɔu	e-wɔ	i-tóa	mbà	mbúcé	ɲaka	è-bòkí	tòò
MBO - SW	n-jɔku	y-ɔkɔ	li-tóka	mbàkà	mbúké	ɲaka	è-wòkí	
KPE - SW	n-jɔku	y-ɔkɔ	li-tóka	mbàkì	mbúké	ɲaka	è-wòkí	tókò
ISU - SW	n-jɔku	y-ɔkɔ	i-tóka	mbàkà	mbúké	ɲaka	è-bòkí	
KOL - SW	n-jɔku	y-ɔkɔ	i-tóka	mbàkà	mbúké	ɲaka	è-bòkí	tókò
BUB - SW	n-jɔku	ey-ɔkɔ	li-tóka	mbàkà	mbúké	ɲaka		
Kundu - N	n-jɔku	ey-ɔkɔ				ɲaka		toko

these sound correspondences does not highlight a conditioning factor that would allow us to predict when we would have /k/, /h/ or Ø for the languages presenting two reflexes. Generally, in cases with similar partially overlapping correspondence sets with no phonological conditioning, the CM dictates that two proto-sounds should be reconstructed (e.g., *k₁ and *h₁). However, a closer examination of the data shows that for the languages with two reflexes, either /h/ ~ /k/ or /Ø/ ~ /k/, the sound sets with /k/ are attested in words that must have been acquired recently (e.g., ‘slave’, ‘cassava’) or words acquired by borrowings (e.g., ‘bag’, ‘sword’). In sum, words that present the sound set with /k/ attested in all the languages are words that cannot be traced back to PB as opposed to the set with /h/ or /Ø/ (e.g., ‘hide’, ‘root’, ‘spear’, ‘rope’, etc.), also attested in a larger number of words. The study of the reflexes /h/ and /Ø/ highlights that PSB *k is undergoing a process of lenition in some Sawabantu languages until its complete deletion in most of the Central languages and some Southwest: PSB *k₁ > /h/ ~ Ø. This type of lenition chain has been described by Pacchiarotti & Bostoen (2022: 12) for West-Coastal Bantu languages spoken in southern Gabon, Congo and Democratic Republic of the Congo with Proto-West-Coastal Bantu *k₂ > x/ɣ/ɛ/h > Ø. In addition, Grollemund & Philippson (to appear), in a paper devoted to the study of the reflexes of PB *k and *g in NW Bantu languages (including A and some B

languages), have observed that PB *k tends to be deleted with the same stages of lenition with generally PB *k₁₂ > /h/ > Ø. The lenition process is confirmed in C2 position with PSB *k₂ being lenited in the same languages (Banoho, Batanga, Du-ala, Bidiman, Oli, Pongo, Mulimba, Mokpe, examples are presented in Table 9). For Bidiman and Oli, we can observe that *k₂ > /ɣ/ (lenition) or Ø whereas in C1, the velar fricative does not appear (because PSB *k₁ has been deleted). In addition, we can notice that PSB *k₁₂ is always maintained and stable in Yasa, Bapuku, Mongo, Mboko, Isu and Kundu. Despite the multiple reflexes observed for PSB *k₁₂, our choice to reconstruct just one proto-sound was probably reasonable as we have observed that it was PSB *k that underwent lenition in some languages (while Yasa, Bapuku, Mongo, Mboko, Isu and Kundu only display /k/) and it is the same PSB *k that was retained in some possibly recently acquired words (that did not have the time to go through the lenition process yet). Finally, having two or more reflexes for a same proto-sound is a phenomenon attested in many Bantu languages with the double reflexes described by Guthrie (1967–1971), Stewart (1973, 1975), Janssens (1991, 1992), Leynseele & Stewart (1980) (see Philippon 2022 for a detailed discussion on double reflexes found in NW Bantu languages) or multiple unconditioned reflexes (MUR) described by Pacchiarotti & Bostoen (2022) referring to a situation where a same proto-sound yields two or more reflexes (without apparent conditioning).

We present in Table 10 the reflexes and proto-sounds reconstructed for voiced plosives. We added the reflexes for PSB *l₁₂ for analysis purposes.

We reconstruct the proto-sound *b₁₂ for the partially overlapping comparative series b/b/β and b/b/w/β (e.g., Table 10) based on the principle of directionality (lenition is more common than fortition, see above). We observe that PSB *b₁₂ > /b/ in South languages, /b/ ~ /β/ in Central languages, /b/ or /β/ in North languages² and PSB *b₁₂ > the fricative /β/ in Southwest languages (lenition) except in Kole. The examples presented in Table 11 and Table 12 show that for Central languages, the distribution of /b/ and /β/ is conditioned. The quality of the following vowel is going to trigger the sound to change: we will find the plosive /b/ before high vowels /_i/ and /_u/ (e.g., Table 11 and Table 12) whereas the implosive /b/ will occur elsewhere (/_a, e, ε, ɔ, o/). This conditioning was already mentioned in Philippon (2022: 18) who demonstrated that PB *b does not present unconditioned double reflexes or MUR in Bantu A languages. For C1 and C2 in Southwest languages except for Kole, when followed by a back vowel (either _o or _u), PSB *b₁₂ will be deleted with sometimes an epenthetic glide /w/ added (e.g., Table 11 and Table 12, examples ‘nine’, ‘snot’, ‘spider’, ‘house’, ‘hippo’).

²Friesen 2002 often proposed the two variants with /b/ or /β/ for a same word.

Table 10: Sound correspondences for voiced plosives and the approximant lateral in C1 and C2 positions. When two or more reflexes are presented, the first one is the most frequent. Bold characters mark sound changes from PSB to the daughter languages.

PSB	*b ₁₂ / _i, u	*b ₁₂ / _a,e,ε,ɔ,o	*b ₁₂ / _u, o	*b ₁ / _wV	*d ₁	*d ₂ / _i, u	*d ₂ / _a,e,ε,ɔ,o	*l ₁₂
BAN - S	b	b	b	b(w)	d	d	d	l
BAT - S	b	b	b	b(w)	d	d	d	l
YAS - S	b	b	b	b(w)	d	d	d	l
BAP - S	b	b	b	b(w)	d	d	d	l
DUA - C	b	b	b	b(w)	d	d	l	l
BID - C	b	b	b	b(w)	d	l	l	l
OLI - C	b	b	b	b(w)	d	l	l	l
PON - C	b	b	b	b(w)	d	d	l	l
MON - C	b	b	b	b(w)	d	d	l	l
MUL - C	b	b	b	b(w)	d	d	d	l
MBO - SW	β	β	w ~∅	gb	l	l	l	l
KPE - SW	β	β	w ~∅	gb	l	l	l	∅ ~l
ISU - SW	β	β	w ~∅	gb	l	l	l	l
KOL - SW	b	b	b	gb	r	r	r	r
BUB - SW	β	β	w ~∅	gb	l	l	l	l ~∅
Kundu - N	b ~β	b ~β	b ~β	gb	r ~l	r ~l	r ~l	r

For the similar partially overlapping comparative sets with d/d/l/r attested in C1 and C2 positions (e.g., Table 13), we chose to reconstruct two proto-sounds *d₁₂ and *l₁₂. The choice was complicated because we can observe the same sets of reflexes for Southwest languages where PSB *d₁₂ and *l₁₂ yield to /l/ (and /r/ in Kole). In addition, the study of the examples presented in Table 13 and Table 14 seems to display a phonological context that would require us to posit one proto-sound (PSB *d or *l) for these comparative sets with the impression that reflexes of *d₁₂ appear before high vowels (_i and _u) whereas the reflexes of *l₁₂ appear elsewhere. And finally, PB *d presenting two reflexes /d/~l/ is a phenomenon that is attested in Bantu languages (Guthrie 1967–1971, Meeussen 1967, Stewart 1973, 1975, 1993, Janssens 1992, Philippson 2022).

In order to untangle this situation, we first need to summarize the reflexes attested for PSB *d₁₂ and *l₁₂ (see Table 13 and Table 14). For South languages, we observe the reflexes /d/ for *d₁₂ and /l/ for *l₁₂. For Central languages, PSB *d₁ > /d/ whereas in C2 position, we can find /d/ or /l/. In Duala, Pongo, Mongo, *d₂ > /d/ when followed by the high vowels /_i, u whereas /l/ is attested elsewhere. For Mulimba, *d₂ > /d/ when followed by high-vowels and /d/ is attested elsewhere.

Table 11: Examples for the reflexes of PSB *b in C1 position

PSB meaning	*b ₁ marriage	*b ₁ two	*b ₁ add	*b ₁ know	*b ₁ _ O, U nine	*b ₁ _ O, U snot	*b ₁ _ O, U spider
BAN - S	dì-ḡáà	ḡé-ḡà	ì-ḡàḡḡ	i-ḡia	dì-ḡùá	dì-ḡòmbó	di-ḡoḡe
BAT - S	dì-ḡáà	ḡé-ḡà	ì-ḡàḡḡ	i-ḡia	dì-ḡùá	dì-ḡòmbó	di-ḡoḡe
YAS - S	ì-ḡáà	ḡí-ḡà	è-ḡàḡḡ		dì-ḡùá	ì-ḡòmbó	di-ḡoḡe
BAP - S	ì-ḡáà	ḡé-ḡà	ì-ḡàḡḡ		dì-ḡùá	dì-ḡòmbó	di-ḡoḡe
DUA - C	dì-ḡáà	ḡé-ḡáà	ḡátà	bía	dì-ḡùá	dì-ḡòmbó	di-ḡoḡe
BID - C	ḡáà	ḡí-ḡáà	ḡárè	bía	ḡùá	ḡòmbó	di-ḡoḡe
OLI - C	ḡáà	ḡí-ḡáà	ḡárè	bía	ḡùá	ḡòmbó	di-ḡoḡe
PON - C	ì-ḡáà	ḡé-ḡáà	ì-ḡátà	i-ḡia	ì-ḡùá	ì-ḡòmbó	i-ḡoḡe
MON - C	dì-ḡáà	ḡé-ḡáà	ì-ḡátà	i-ḡia	dì-ḡùká	dì-ḡòmbó	di-ḡoḡe
MUL - C	dì-ḡáà	ḡé-ḡáà	ì-ḡátà	i-ḡia	dì-ḡùá	dì-ḡòmbó	di-ḡoḡe
MBO - SW	li-ḡáà	ḡé-ḡáà	li-ḡátà		lù-ùká	wòmbó	li-woḡe
KPE - SW	li-ḡáà	ḡé-ḡáà	li-ḡátà	li-ḡia	lù-ùwá	li-òmbó	li-woḡe
ISU - SW	li-ḡáà	ḡé-ḡáà	li-ḡátà	i-ḡia	li-ḡùká	li-wòmbó	li-woḡe
KOL - SW	dì-ḡáà	ḡé-ḡáà	ì-ḡátà	i-ḡia	dì-ḡùá	dì-ḡòmbó	di-ḡoḡe
BUB - SW	li-ḡáà	ḡé-ḡáà	li-ḡátà	li-ḡia	li-ḡbá	li-òmbó	li-woḡe
Kundu - N		be-be/ ḡe-ḡe					ri-ḡoḡe

Table 12: Examples for the reflexes of PSB *b in C2 position

PSB meaning	*b ₂ six	*b ₂ divide	*b ₂ banana	*b ₂ wickedness	*b ₂ / _o, u house	*b ₂ / _o, u hippo
BAN - S	n-tóḡá	i-kaḡa	di-húḡé	ḡoḡé	ndáḡo	ḡḡubú
BAT - S	n-tóḡá	i-kaḡa	di-húḡé	ḡoḡé	ndáḡo	ḡḡubú
YAS - S	mo-tóóḡá	e-kaḡa	vi-kúḡé			ḡḡubú
BAP - S	n-tóḡá	i-kaḡa	i-kúḡé	ḡoḡé	ndáḡo	ḡḡubú
DUA - C	mu-tóḡá	j-aḡa	di-kúḡé	ḡoḡé	ndáḡo	ḡḡubú
BID - C	n-tóḡá	y-aḡa		ḡoḡi	ndáḡo	ḡḡubú
OLI - C	n-tóḡá	y-aḡa		ḡoḡi	ndáḡo	ḡḡubú
PON - C	mu-tóḡá	ij-aḡa	i-kúḡé	ḡoḡé	ndáḡo	ḡḡubú
MON - C	mu-tóḡá	-kaḡane	e-kúḡékúḡé	ḡoḡée	ndáḡo	ḡḡubú
MUL - C	mu-tóḡá	i-kaḡa	du-úḡé	ḡoḡé	ndáḡo	ḡḡubú
MBO - SW	mo-tóḡá			wóḡée	ndáwo	ḡḡuwú
KPE - SW	mo-tóḡá			wóḡée	ndáwo	ḡḡuwú
ISU - SW	mo-tóḡá	i-kaḡa		wóḡée	ndáwo	ḡḡuwú
KOL - SW	mo-tóḡá			ḡoḡé	ndábo	ḡḡubú
BUB - SW	mo-tóḡá	li-kaḡa		wóḡée	ndáwo	ḡḡuwú
Kundu - N		-kaḡa			ndabo/ndaḡo	

Table 13: Examples for the reflexes of PSB *d and *l in C1 position

PSB meaning	*d ₁ extinguish	*d ₁ old	*d ₁ water	*d ₁ run	*l ₁ send	*l ₁ read	*l ₂ three
BAN - S	i-ďima	i-ďùnà	ma-ďibá	i-ďàngwà	i-lóma	i-làngà	ḡé-lálo
BAT - S	i-ďima	i-ďùnà	ma-ďibá	i-ďàngwà	i-lóma	i-làngà	ḡé-lálo
YAS - S	e-ďima	è-ďùnà	me-héḡá		è-lóma	è-làngà	ḡi-lálo
BAP - S	i-ďima	i-ďùnà	mi-iḡa		i-lóma	i-làngà	ḡé-lálo
DUA - C	ďima	ďùnà	ma-ďibá	ďàngwà	è-lóma	làngà	ḡé-lálo
BID - C	ďime	ďùnà	man-ďiba		è-lóma	làngà	ḡi-lálo
OLI - C	ďime	ďùnà	ma-ďibá		è-lóma	láyà	ḡi-lálo
PON - C	i-ďima	i-ďùnà	ma-ďibá	i-ďàngwà	i-lóma	i-làngà	ḡé-lálo
MON - C	i-ďima	i-ďùnà	ma-ďibá		i-lóma	i-làngà	ḡé-lálo
MUL - C	i-ďima	i-ďùnà	ma-ďibá	i-ďàngwà	i-lóma	i-làngà	ḡé-lálo
MBO - SW	li-lima	lù-ùná	ma-ďiḡá		li-lómà	li-làngà	ḡé-lálo
KPE - SW	li-lima	lù-ùná	ma-ďiḡá		lò-óma	li-làngà	ḡé-yálo
ISU - SW	i-lima	i-ďùnà	ma-ďiḡá		li-lómà	i-làngà	ḡé-lálo
KOL - SW	i-ďima	i-ďùnà	ma-riḡá		i-róma	i-ràngà	ḡé-láro
BUB - SW	li-lime	li-lùná	ma-ďiḡá		li-yóma	li-yàngà	ḡé-lálo
Kundu - N	rime	unu	i-liba/i-riḡa		loma/roma		be-raro

Table 14: Examples for the reflexes of PSB *d and *l in C2 position

PSB meaning	*d ₂ turtle	*d ₂ birth	*d ₂ rope	*d ₂ goat	*d ₂ put inside	*l ₂ lend	*l ₂ write	*l ₂ beard
BAN - S	kúḡù	i-yáḡi	ḡ-hoḡi		i-wéḡè	i-ḡálè	i-tila	ḡi-heḡú
BAT - S	kúḡù	i-yáḡi	ḡ-hoḡi	mbóḡi	i-véḡè	i-ḡálè	i-tila	ḡi-heḡú
YAS - S	kúḡù	i-jáḡi	mo-kóḡi		è-véḡè	è-ḡálè	e-tila	seḡu
BAP - S	kúḡù	i-yáḡi	ḡ-kóḡi		i-wéḡè	i-ḡálè	i-tila	i-heḡú
DUA - C	wúḡù	yáḡi	m-ḡḡi	mbóḡi	wéḡè	ḡálè	tila	seḡú
BID - C	wúlù	yálè	m-ḡli	mbóli	féḡè	ḡálè	tila	seḡú
OLI - C	wúlù	yáli	m-ḡli	mbóli	wéḡè	ḡálè	tila	seḡú
PON - C	wúḡù	jadi	m-ḡḡi	mbóḡi	i-wéḡè	i-ḡálè	i-tila	i-seḡú
MON - C	kúḡù	jadi	mu-kóḡi	mbóḡi	i-wéḡè	i-ḡálè	i-tila	i-seḡú
MUL - C	kúḡù	i-yáḡi	m-ḡḡi	mbóḡi	i-éḡè	i-ḡálè	i-tila	ḡi-heḡú
MBO - SW	è-kúḡè	li-yáli	mo-kóli	mbóli	li-wálè	li-ḡálè	li-tila	nḡeli
KPE - SW	kúlù	li-yáli	m-ḡli	mbóli		li-ḡálè	li-tila	nḡelu
ISU - SW	è-kúḡè	jali	mo-kóli	mbóli		i-ḡálè	i-tila	i-seḡú
KOL - SW	è-kúḡè	jáḡi	mo-kóri	mbóri	i-wéḡè	i-báḡè	i-tira	i-sèḡú
BUB - SW	è-kúḡè	li-yáli	m-ḡli	mbóli	li-wálè	li-ḡálè	li-tila	
Kundu - N	ku	ja	mo-kori	mboli/mbori	ule		tira	

For Bidiman and Oli, $*d_2 > /l/$ in all contexts. For PSB $*l_{12}$, we found the reflexes $/l/$ for these languages. For Southwest languages, the situation is simpler with $*d_{12} > /l/$ and $*l_{12} > /l/$ in all the languages except for Kole that presents $/r/$ for both proto-sounds. We can note that Mokpe and Bubia can present the reflexes $\emptyset$ or $/l/$ for PSB $*l_{12}$. Finally, in Kundu (North), the data presented in Friesen 2002: 24 gave several reflexes with $*d_{12} > /l/$ or $/r/$ (Friesen 2002: 25-27 explained that there is one phoneme $/d/$ in North languages that is realized $[d]$ in front of the high vowels $/i/$ and $/u/$ whereas $[l]$ is attested elsewhere with $[r]$ occurring rarely). And $*l_{12} > /r/$ in Kundu.

If we posit one proto-sound such as PSB $*d_{12}$ or $*l_{12}$ with the rule (conditioning) that the reflexes $/d/$ or $/d'/$ are attested in Sawabantu languages before high vowels whereas $/l/$ will be attested elsewhere, we would face several issues for the South and Central languages. First, for South languages, while we have observed that $/d'/$ is more often attested in front of high vowels whereas $/l/$ is more attested in front of non-high vowels, reconstructing one proto-sound would oblige us to posit double reflexes or MUR in C1 and C2 positions as $/d/$ and $/l/$ can appear in similar phonological contexts (e.g., Table 13 examples ‘run’ and ‘read’; Table 15, examples ‘dig’, ‘be full’, ‘put in’, ‘lend’ or ‘oil’). Second, for Central languages, in C1 position, reconstructing one proto-sound is tempting as we found the reflexes $/d/$ only before high vowels and $/l/$ elsewhere. However, the situation in C2 with

Table 15: Comparison of PSB $*d_2$ and $*l_2$.

PSB meaning	*bodi goat	*kado ladder	*bolo work	*pud dig	*ul be full	*ved put in	*bal lend	*ula oil
BAN - S		bò-hàḁò		i-puḁa	ì-húlà	ì-wéḁḁ	ì-bálè	mà-úlè
BAT - S	mbóḁi	bò-hàḁò		i-puḁa	ì-húlà	ì-véḁḁ	ì-bálè	mà-húlè
YAS - S			è-bóló	e-puḁwa	è-hólà	è-vélè	è-bálè	màv-úlè
BAP - S		bò-kàḁò		i-puḁa	ì-húlà	ì-wéḁḁ	ì-bálè	má-lè
DUA - C	mbóḁi		è-bóló	pula	júlà	wélè	bálè	mù-úlà
BID - C	mbóli		bóló		ólà	félè	bálè	mù-úlà
OLI - C	mbóli		bóló		ólà	wélè	bálè	mù-úlà
PON - C	mbóḁi		è-bóló	i-pula	ì-júlà	ì-wélè	ì-bálè	mù-úlà
MON - C	mbóḁi		è-bóló	i-pula	ì-júlà	ì-wélè	ì-bálè	mà-wúlà
MUL - C	mbóḁi	bò-kàḁò	è-bóló	i-puḁa	iúlà	iéḁḁ	ì-bálè	mù-úlè
MBO - SW	mbóli		è-wóló		lù-úlà	li-wálè	li-βálè	màl-úwà
KPE - SW	mbóli		è-wóló		lù-újà		li-βálè	mà-újà
ISU - SW	mbóli		è-wóló	i-ḁulwa	yù-úlà		ì-bálè	mà-úlà
KOL - SW	mbóri		è-wóró	i-ḁula	i-wúrà	ì-wérè	ì-bàrè	mà-úrà
BUB - SW	mbóli		è-wóló	li-ḁua		li-wálè	li-bálè	
Kundu - N	mboli					ule		

the conditioning described and more specifically the three reflexes observed in Mulimba with /d/, /d/ and /l/ tipped the scales in favor of the reconstruction of two proto-sounds. In Mulimba, /d/ is found in front of high vowels (same as Duala, Pongo and Mongo) but /d/ (as opposed to /l/ attested in Duala, Pongo and Mongo) is found in front of non-high vowels whereas /l/ is attested in many contexts, mostly in front of non-high vowels. The conditioning observed for /d/ and /d/ along with the similar contexts in which we can find /d/ and /l/ in Mulimba shows that these sets of reflexes should be linked to two separate proto-sounds PSB *d and *l. An examination of Table 15 (along with Table 13 and Table 14) confirms the need of two distinct proto-sounds which resulted in two sound sets: /d/ /d/ (< PSB *d) are attested in Mulimba when South languages also present /d/ and reflexes /l/ (< PSB *l) are attested in Mulimba when South languages also present /l/.

Finally, there is some uncertainty as to the phonetic nature of the PSB (and PB) *b and *d as we cannot be absolutely sure that the original sounds were not implosives like in Southern and were later strengthened in some languages in front of first-degree vowels.

We do not reconstruct *g in PSB as the phoneme /g/ is absent in C1 and C2 positions in the daughter languages (except when preceded by a nasal homorganic prefix of class 9 such as in “moon” PSB *ŋ-gɔnde). This phenomenon is not unusual as it was noticed earlier by Möhlig (1981) and Nurse & Philippson (2003) who reported that Bantu languages spoken in the rainforest (generally A, B, C and parts of H) present this particular sound change where PB *g > /k/. In addition, Pacchiarotti & Bostoen (2020) showed that for West-Coastal languages (parts of B and H), there was a velar merger that led PB *k and *g to merge into /k/. Finally, Grollemund & Philippson (to appear) described that for NW Bantu languages (A and some B20), PB *k > Ø which triggered the PB *g > /k/. These sound changes modified the phonemic inventory of these Bantu languages yielding to the disappearance of /g/ in some languages.

We also chose not to reconstruct labial-velar stops consonants *kp and *gb attested in Southwest and North Sawabantu languages (e.g., Table 5 and Table 10) as they are irregular and generally the result of contact or the fortition of the PSB sequences *kwV and PSB *bwV. Labial-velar stops are rare in Bantu languages but they do exist in several Bantu languages belonging to zones A, B and C. It is assumed that Bantu languages have acquired these labial-velar stops through contact with either Ubangian, Central Sudanic or Southern Bantoid languages (Clements & Rialland 2008, Bostoen & Donzo 2013, Idiatov & Van de Velde 2021). Due the geographic location of Sawabantu languages in southern Cameroon, we could assume that Southwest and North Sawabantu languages have acquired

these labial-velar stops by prolonged contact with Southern Bantoid languages located in the same region. While this hypothesis is highly plausible (and only a thorough study of these labial-velar stops will give us a clear answer), we have noticed some conditioning environments that might explain the presence of some labial-velar stops in Sawabantu languages. Bostoen & Donzo (2013: 450) examine the origin of labial-velar stops in Ngombe (C41), a Bantu language spoken in Democratic Republic of the Congo. While they do concur on the fact that labial-velar stops have been acquired by contact with neighboring Ubangian speakers, they also mention that conditioned sound changes can trigger labial-velar stops to develop. They demonstrate that labial-velars in Ngombe that do not have a Ubangian origin have developed because of the following rule: $C_{[+voiced]}wV > gbV$ and $C_{[-voiced]}wV > kpV$, proposed first by Connell (1994), Connell (1998) and Cahill (1999). It seems that we have a similar phenomenon in Southwest and North Sawabantu languages where some labial-velar stops have possibly been acquired through contact while others seem to obey to this same conditioned sound change presented in Table 16. The status of these labial-velars stops /kp/ and /gb/ is still unclear and more data need to be analyzed.

Table 16: Examples of the labial-velar stops in Southwest and North groups.

PSB	South	Central	Southwest	North
*kwale “partridge”	/ŋgwae/	/kwale/	/kpae/	
*kwedi “death”	/wedī/	/kweli/	/kpeli/	
*bwa “dog”	/mbwa/	/mbwa/	/ŋgba/	/mgbá/

For the fricatives presented in Table 17, we have reconstructed *v and *s in C1 and C2 positions. For the correspondence set with v/w/f/Ø, we chose to reconstruct *v since weakening (*v > /w/ > Ø) is more likely than strengthening. We could also have proposed *f but we hesitated to do so on the basis of Bidiman alone, which is rather idiosyncratic on other counts. For the set with h/s/z, we reconstructed *s in C1 and C2 positions based on the same principle (with PSB *s₁₂ undergoing lenition to /h/ in South languages and also Mulimba). The exact phonetic value of the Mokpe sound /z/ is the subject of some debate in the literature (Atindogbe 2013: 9).

Nasal consonants *m, *n, *ɲ have been reconstructed in C1 and C2 position along with prenasalized consonants *mb, *nd, *ŋg (e.g., Table 19).

Table 17: Sound correspondences for fricatives in C1 and C2 positions. Bold characters mark sound changes from PSB to the daughter languages.

PSB	*v ₁₂	*s ₁₂	PSB	*v ₁₂	*s ₁₂
BAN - S	v	h	MON - C	w	s
BAT - S	v	h	MUL - C	Ø	h
YAS - S	v	h	MBO - SW	Ø	s
BAP - S	v	h	KPE - SW	Ø	ʒ
DUA - C	w	s	ISU - SW	Ø	s
BID - C	f	s	KOL - SW	w	s
OLI - C	v	s	BUB - SW	w	s
PON - C	w	s	Kundu - N	Ø	s

Table 18: Examples for the reflexes of PSB *v and *s in C1 and C2 positions.

PSB meaning	*v ₁ wind	*v ₁ twin	*v ₂ itch	*s ₁ iron	*s ₁ army	*s ₂ corn	*s ₂ choose
BAN - S	dī-vîndî	dī-váhà	yòvá	i-hîa	há o	mbáyí	
BAT - S	dī-vîndî	dī-váhà	yòvá	i-hîa		mbáhi	i-pòho
YAS - S	i-vîndó	ì-vágà	òvá	e-hîsa	i-yáko	mbàsi	
BAP - S	dī-vîndî	ì-váhà	yòvá	i-hîa	há o	mbáyí	
DUA - C	di-wîndî	dì-wásà		sîa	sáo	mbàsi	pɔsɔ
BID - C	findîfindî	fàsà		sîa	sáo	mbàsi	pɔsɔ
OLI - C	vîndîvîndî	vàsà		sîa	sáo	mbàsi	pɔsɔ
PON - C	i-wîndîwîndî	ì-wásà		i-sîa	sáo	mbàsi	i-pɔsɔ
MON - C	di-wîndîwîndî	dì-wásà		i-sîa	sáko	mbàsi	i-pɔsɔ
MUL - C	di-îndî	dì-áhà	yòwá	i-hîa	há o	mbàsi	i-pòho
MBO - SW	li-wîndó	lì-ásè		li-sîa	sáko		li-ɸɔsɔ
KPE - SW	li-wîndî	mà-ézè		li-ʒîa	ʒiyaó	mbàʒi	li-ɸɔʒɔ
ISU - SW	li-wîndîwîndî	mà-ésè		i-sîa	sáo	mbàsi	li-ɸɔsɔ
KOL - SW	di-wîndî	dì-wásà		i-sîa	síako	mbàsi	li-ɸɔsɔ
BUB - SW	li-îndó	m-ásè		li-sîa	síyaó	mbàsi	li-ɸɔsɔ
Kundu - N		ri-ase					βɔsɔ

Table 19: Sound correspondences for nasals and prenasalized consonants. Bold characters mark sound changes from PSB to the daughter languages. We can note that denasalization is frequent in Oli.

PSB	* m ₁₂	* n ₁₂	* ɲ ₁₂	* mb	* nd	* ɲg
BAN - S	m	n	ɲ	mb	nd	ɲg
BAT - S	m	n	ɲ	mb	nd	ɲg
YAS - S	m	n	ɲ	mb	nd	ɲg
BAP - S	m	n	ɲ	mb	nd	ɲg
DUA - C	m	n	ɲ	mb	nd	ɲg
BID - C	m	n	ɲ	mb	nd	ɲg
OLI - C	m	n	ɲ	mb	nd	ɣ ~ɲg
PON - C	m	n	ɲ	mb	nd	ɲg
MON - C	m	n	ɲ	mb	nd	ɲg
MUL - C	m	n	ɲ	mb	nd	ɲg
MBO - SW	m	n	ɲ	mb	nd	ɲg
KPE - SW	m	n	ɲ	mb	nd	ɲg
ISU - SW	m	n	ɲ	mb	nd	ɲg
KOL - SW	m	n	ɲ	mb	nd	ɲg
BUB - SW	m	n	ɲ	mb	nd	ɲg
Kundu - N	m	n	ɲ	mb	nd	ɲg

Table 20: Examples for the reflexes of PSB *m, *n and *ɲ in C1 and C2 positions.

PSB meaning	*m ₁ finish	*m ₂ stand	*n ₁₂ bird	*n ₂ shame	*ɲ ₁ surprised	*ɲ ₁ happiness	*ɲ ₂ shine
BAN - S	ì-mà	i-témɛ	i-nɔní	ì-yónì		e-ɲɛɲɛ	ì-pàɲà
BAT - S	ì-mà	i-témɛ	i-nɔní	ì-hónì		e-ɲɛɲɛ	ì-pàɲà
YAS - S		e-témɛ	vi-ɲɔní	vi-yónì		e-ɲɛɲɛ	è-pàɲà
BAP - S	ì-mà	i-témɛ	i-nɔní	ì-yónì		e-ɲɛɲɛ	ì-pàɲà
DUA - C		témɛ	i-nɔón	ì-sóòn	ɲáka	mu-ɲɛɲɛ	pàɲa
BID - C	mà	témɛ	nɔón	sóòn	ɲáyá	ɲì-ɲɛɲɛ	pà
OLI - C	mà	témɛ	nɔón	sóòn	ɲáyá	ɲì-ɲɛɲɛ	pà
PON - C		i-témɛ	e-nɔón	è-sóòn	i-ɲáka	mu-ɲɛɲɛ	ì-pàɲà
MON - C	ì-mà	i-témɛ	i-nɔón	ì-sóòn	i-ɲáka	mu-ɲɛɲɛ	ì-pàɲà
MUL - C		i-témɛ	i-nɔón	ì-hóòn	i-ɲáka	mu-ɲɛɲɛ	ì-pàɲà
MBO - SW		li-témɛ	i-nɔní	ì-sónì	li-ɲáka	mo-ɲɛɲɛ	li-kpàɲà
KPE - SW		li-témɛ	i-nɔní	ì-zónì	li-ɲáka	mo-ɲɛɲɛ	
ISU - SW	ì-mà	i-témɛ	i-nɔní	ì-sóòn	i-ɲáka	mo-ɲɛɲɛ	ì-pàɲà
KOL - SW	ì-mà	i-témɛ	i-nɔní	ì-sónì	i-ɲáka	mo-ɲɛɲɛ	ì-pàɲà
BUB - SW		li-témɛ	i-nɔní	ì-sónì	li-ɲáka	mo-ɲɛɲɛ	
Kundu - N	mare		i-no	i-so	ma-ɲaka		

Prenasalized consonants can also be found in C1 position as they are the result of the combination of the nasal prefix of class 9 *N- (homorganic nasal) combined with a plosive consonant. For example, the word for ‘crocodile’ is reconstructed *gàndó in PB (BLR3 1326) with a class 9 which led to the PB root *N-gàndó > [ŋ-gàndó] in Banoho.

Table 21: Examples for the reflexes of PSB *mb, *nd and *ŋg.

PSB meaning	*mb namesake	*nd crocodile	*ŋg root
BAN - S	mbómbò	ngàndó	ŋ-hàngá
BAT - S	mbómbò	ngàndó	ŋ-hàngá
YAS - S	mbómbò	ngàndó	
BAP - S	mbómbò	ngàndó	ŋ-kàngá
DUA - C	mbómbò	ngàndó	mw-àngá
BID - C	mbómbò	ngàndó	mw-àngá
OLI - C	mbómbò	ngàndó	mw-àŋá
PON - C	mbómbò	ngàndó	mw-àngá
MON - C	mbómbò	ngàndó	mù-kàngá
MUL - C	mbómbò	ngàndó	mw-àngá
MBO - SW	mbómbò	ngàndó	ŋm-àngá
KPE - SW	mbómbò	ngàndó	ŋm-àngá
ISU - SW	mbómbò	ngàndó	mò-kàngá
KOL - SW	mbómbò	ngàndó	ŋw-àngá
BUB - SW	mbómbò	ngàndó	ŋm-àngá
Kundu - N		ngando	

We also reconstructed the approximants *w and *y (presented in Table 22) in C1 and C2 positions.

The study of *w₁₂ and *y₁₂ shows that in many examples, /w/ and /y/ are epenthetic consonants in Sawabantu languages resulting from various processes occurring at morpheme boundaries. Generally, when PSB *k deletes, it leads to the addition of /w/ or /y/ (depending on the quality of the following vowel) in between the prefix and the root: PSB *kuba ‘hen’ > /wuḃa/ in Duala, Bidiman, Oli and Pongo. Also, when PSB *b > Ø / _o, u, it leads to the addition of /w/: PSB *bómbò ‘snot’ > /wómbò/ in Mbole. Finally, we also have examples where PSB *v > /w/ in some languages: PSB *vasa ‘twin’ > /wasa/ in Pongo. However, there are also cases where *w can be reconstructed in C1 and C2 position (e.g., Table 23) or

Table 22: Sound correspondences for approximants *w and *y.

PSB	*w	*y	PSB	*w	*y
BAN - S	w	y	MON - C	w	y
BAT - S	w	y	MUL - C	w	y
YAS - S	w	y	MBO - SW	w	y
BAP - S	w	y	KPE - SW	w	y
DUA - C	w	y	ISU - SW	w	y
BID - C	w	y	KOL - SW	w	y
OLI - C	w	y	BUB - SW	w	y
PON - C	w	y	Kundu - N	w	y

in between C1 and V1 (e.g., PSB *bwa ‘dog’ > /imbwa/ in Batanga for example) or C2 and V2 (e.g., PSB *pungwa ‘mix’ > /i-pungwa/ in Banoko for example). Finally, we needed to reconstruct *y₁ for comparative series where /y/ is attested in all Sawabantu languages (e.g., Table 23, examples PSB *ya ‘give birth’ or PSB *yoko ‘penis’). We did not find non-epenthetic *y in C2 position.

We do find in Ebobissé’s data (2015) the occurrence of /j/ (/ɟ/) that appears rarely and for this reason, we have decided to not reconstruct *j as it is often a prefix attached to a root that starts with a vowel (e.g., Table 24 ‘ten’ or ‘mirror’) or in free variation with /y/ for which we have reconstructed *y (e.g., Table 24 “wash”).

We reconstruct in total 16 PSB consonants: *p, *t, *k, *b, *d, *l, *m, *n, *ɲ, *v, *s, *w, *y, *mb, *nd, *ŋg. The PB reconstructed by Guthrie (1967–1971) and Meeussen (1967) are very similar; although they include the addition of PB *g and *c (Guthrie also distinguished *j from *y, see Wills 2022 for details). In these PB system reconstructions, there are no fricative consonants reconstructed (see Bostoen 2019 for more details on PB).

Hedinger (1987) reconstructed for PM in C1 and C2 positions almost the same set of consonants with *p, *t, *k, *s, *b, *d, *j, *m, *n, *ny, *mb, *nd, *ŋg and *w. He also had *f (but not *v) and he reconstructed *g. A closer look at Hedinger’s data showed that the reflexes of PM *g in C1 position are only attested when preceded by a nasal prefix (Hedinger 1987: 39). However, in C2, we observe that PM *g is lenited in some languages displaying either /ʔ/, /kʔ/ or /gʔ/. The choice made by Hedinger to reconstruct *g in PM is understandable as we find attestations of /g/ in C1 and C2 positions as opposed to Sawabantu languages where /g/ is absent in C2 (for example, we have in PM ‘penis’ *jòg’ > [n-jògʔ] in Akossi whereas we have *yoko in PSB).

Table 23: Examples for the reflexes of PSB *w and *y in C1 and C2 positions.

PSB meaning	*w ₁ die	*w ₂ coast	*y ₁ penis	*y ₁ give birth
BAN - S	i-wóó		yòó	i-yáa
BAT - S	i-wáa		yòó	i-yáa
YAS - S	e-wáa			e-jáa
BAP - S	i-wáa		yòó	i-yáa
DUA - C	wó	sáwá	è-yòó	yáa
BID - C	wáa	sáwá	yòó	yáa
OLI - C	wáa	sáwá	yòó	yáa
PON - C	ij-óó	sáwá	è-yòó	i-yáa
MON - C	i-wáa	sáwá	è-yòkó	i-yáa
MUL - C	i-wó	sáwá	è-yòó	i-yáa
MBO - SW	li-wáa	sáwá	yòkó	li-jáa
KPE - SW	liŋm-áa	záwá	yòkó	li-yáa
ISU - SW	i-wáa	sáwá	yòkó	i-yáa
KOL - SW	i-wáa	sáwá	è-yòkó	i-yáa
BUB - SW	li-wáa	sáwá	è-yòkó	li-yéé
Kundu - N	wa			

The comparison of the two proto-systems reconstructed for Manenguba and Sawabantu (languages spoken in the same region, e.g., Figure 1 and classified within the same subgroup, e.g., Figure 2) illustrates some similarities between the two subgroups; however, we also observe some key differences. For example, the same sets of consonants have been reconstructed in both proto-languages (with the exception of *g and *v) but the study of the reflexes within the daughter languages shows different evolution patterns for the same proto-consonants in PM and PSB. The examination of Table 25 displaying the reflexes of PM's consonants (without the nasals) in C1 position shows that Manenguba languages seem more conservative (with respect to the protolanguage) than Sawabantu languages where multiple reflexes are attested for the same reconstructed consonants (e.g., PSB *d > /d/, /d/, /l/ or /r/). However, the loss of *V2 in Manenguba languages led to the weakening of *C2 triggering the loss of the voicing distinction for plosives whereas in Sawabantu languages, V2 is maintained (along with the voicing distinction). Finally, vowels in Manenguba languages underwent many changes (Hedinger 1987: 44-45) as opposed to Sawabantu languages.

Table 24: Examples with /j/ in Sawabantu languages.

PSB meaning	- ten	- mirror	*y ₁ wash
BAN - S	jóòm	jɛní	i-yoa
BAT - S	jóòm	jɛní	i-yoa
YAS - S	jóòm	jɛní	
BAP - S	jóòm	jɛní	i-yoa
DUA - C	d-óòm	jɛné	jóa
BID - C	jóòm	jɛné	yoa
OLI - C	jóòm	jɛné	yoa
PON - C	d-óòm	jɛné	i-jóa
MON - C		jɛné	i-jóa
MUL - C	jóòm	jɛné	i-yoa
MBO - SW	dì-ómè		
KPE - SW	lì-yómè	li-ɛní	
ISU - SW		jɛní	i-yoa
KOL - SW	jómì	jɛné	yoa
BUB - SW	dì-ómì	li-ɛní	
Kundu - N			

3.2 Sound changes in Sawabantu languages and preliminary observations

The sound changes observed in Table 5, Table 10, Table 17, Table 19, and Table 22 have shown that Sawabantu languages underwent many changes with several PSB consonants undergoing lenition. The analysis of sound changes supports three genetic subgroups in accordance with the three geographical subgroups distinguished in Ebobissé (1989) and Ebobissé (2015) with the addition of Kundu (North) with (i) South: PSB *b₁₂ > /b/, PSB *d₁₂ > /d/ and PSB *s₁₂ > /h/, (ii) Central: we have a phonological conditioning found for PSB *b₁₂ and PSB *d₂ with also PSB *k₁₂ > Ø in Duala, Bidiman, Oli, Pongo and Mulimba, and (iii) Southwest and North: PSB *d₁₂ > /l/, the lenition of PSB *p₁₂ > /ɸ/ and PSB *b₁₂ > /β/. Therefore, the definition of our sub-groups differs from Ebobissé's (1989) analysis in so far as he did not include Oroko (aka North) and so did not notice the proximity between the latter and Southwest.

In addition, the treatment of PSB *d₁₂ (and *l₁₂) is paramount in Sawabantu languages for possibly establishing some putative chronology. With the three ge-

Table 25: Reflexes of PM's plosives, fricatives and the approximant lateral in C1 position (Hedinger 1989: 38-39).

PM	*p ₁	*t ₁	*k ₁	*b ₁	*d ₁	*g ₁	*s ₁	*f ₁	*l ₁
MBM	p	t	k	b	d	g	s	f / h	l / d
MBM	p	t	k	b	d	g	s / ʃ	f	l / d
MYE	p	t	k	b / bʰ	d	g	s	h	l
MBE	p	t	k	b	d	g	s / ʃ	h	l
ELU	p	t	k	b	d	g	s	h	l
NNE	p	t	k	b	d	g	s	h	l
AKO	p	t	k	b	d	g	s	h	l
MHE	p	t	k	b	d	g	s	h	l
MWK	p	t	k	b	d	g	s	h	l
MKA	p	t	k	b	d	g	s	h	l
BLN	p	t	k	b	d	g	s	h	l
BBO	p	t	k	b	d	g	s	h / f / v	y
LEF	p	t	k	b	d	g	s / ʃ	h	l / d
LEK	p	t	k	b / p	d	g	s	h	l / d

netic subgroups distinguished (South, Central and Southwest+North), how PSB *d₁₂ evolved in these languages is informative. For Southwest languages, PSB *d₁₂ and *l₁₂ have merged into /l/. This phonological merger defining Southwest and North languages must have occurred last because otherwise, all the PSB *d would be realized /l/ in the remaining subgroups. Instead, the distinction between PSB *d₁₂ and *l₁₂ is maintained in Central and South languages. The retention of PSB *d₁ in Central languages along with the conditioning observed in some Central languages for PSB *d₂ as opposed to the unconditioned sound change PSB *d₁₂ > /d/ in South languages confirms that we are dealing with two different subgroups.

Finally, the results observed are in accordance with the groupings distinguished in lexicon-based phylogenies with South and Central languages classified under a same node whereas Southwest and North are classified in a different subgroup linked to Manenguba languages (e.g., Figure 2).

3.3 Comparing Proto-Bantu and Proto-Sawabantu: new perspectives

In this section, we compared our reconstructions established for PSB with PB reconstructions in order to establish sound correspondences between the two

proto-languages with a focus on plosive consonants in C1 and C2 positions. The aim is to bring a new perspective on the phenomena described in the previous section by studying Sawabantu languages in relation to Proto-Bantu.

The examination of Table 26 shows that PB *p yielded to two reflexes in PSB with *p and *v. Philippson 2022: 25-26 mentioned a possible partial conditioning *pVp that could trigger the strong reflexes of PB *p for NW Bantu languages which prevented him from positing double reflexes for PB *p. For PB *t, we have a case of double reflexes or MUR where PB *t > PSB *t or *l with no apparent conditioning. However, Philippson (2022: 30) did notice that the strong reflexes of PB *t tend to appear before high vowels in several NW Bantu languages. Our data shows that the reflexes of PSB *t and *l occur in the same phonological contexts confirming the status of double reflexes for PB *t with however one observation: the sequences /liC2V2/, /luC2V2/ or /C1V1li/, /C1V1lu/ are extremely rare in Sawabantu languages. As mentioned in the previous section, PB *k > Ø in Sawabantu languages but also in the remaining NW Bantu languages (A and some B20) which triggered PB *g > *k in the same languages (Grollemund & Philippson to appear). PB *b is retained in PSB. Finally, PB *d is deleted when followed by non-high vowels and retained (PSB *d) when followed by the vowels *i and *u, a phenomenon observed for many Bantu A languages (Philippson 2022: 30-31).

The comparison between PB and PSB shed a new light on the origin of the phoneme /l/ in Sawabantu languages. For Southwest languages, the reflexes /l/ originated from two different PB consonants: *t and *d (reconstructed *l and *d in PSB) which confirms why we had two distinct sound sets in Section 3.1.2. It also explains why the reflexes of PSB *d can only be found before high vowels as PB *d > Ø before non-high vowels. According to Philippson (2022: 34), the deletion of PB *d before non-high vowels occurred with an intermediate stage attested in some NW Bantu languages with PB *d > /l/ > Ø (whereas PB *d > /d/ or /d/ before high vowels). We found this intermediate stage in some Central Sawabantu languages where we have observed for C2 PSB *d > /l/ before non-high vowels confirming this lenition chain which implied that PB *d₂ > /l/ or /d/ or Ø before non-high vowels.

In addition, according to Philippson (2022: 37), the sound change PB *d > (*l) > Ø must have occurred before PB *t > PSB *l otherwise, most of the /l/ reflexes of PSB *l would have been deleted. Therefore, it seems that the loss of PB *d (> weak reflexes) triggered PB *t > PSB *l (as well as in some other NW Bantu languages, see Philippson 2022: 30-31) with the result of reintroducing the missing sound /l/. Interestingly, in Sawabantu languages, the reflexes of PSB *l are found in front of non-high vowels which could corroborate that the sound change from PB *t

Table 26: Comparison between PB and PSB for plosives in C1 and C2 positions.

PB	PSB	S	C	SW	N	Examples in PB (from BLR3)	PSB
*p	*p	p	p	ϕ	ϕ	PB *pàpà “wing” (2407)	PSB *papa
	*v	v	w ~v ~f	Ø ~w	Ø	PB *pàcà “twin” (2348)	PSB *vasa
*b	*b	ɓ	b	β	b ~β	PB *cùpà “calabash” (746) PB *bàdɪ “two” (9093)	PSB *sova PSB *baa
	*t	t	t	t	t	PB *tùóbà “six” (3034) PB *tòì “ear” (3030)	PSB *toba PSB *toi
*t						PB *dùt “pull” (1267)	PSB *dut
	*l	l	l	l ~r	r	PB *téndé “palm tree” (2849)	PSB *lende
*d						PB *tátò “three” (2811) PB *dòmè “husband” (1183)	PSB *lalo PSB *ome
	Ø	Ø	Ø	Ø	Ø	PB *dànd “to buy” (849)	PSB *and
*d						PB *bùdà “rain” (368)	PSB *bua
						PB *dibà “water” (1025)	PSB *diba
*k	*d	d	d ~l ~d	l ~r	r	PB *dùt “pull” (1267)	PSB *dut
						PB *kùdù “tortoise” (2110) PB *kádí “wife” (1674)	PSB *kudu PSB *adi
*g	Ø	Ø	Ø	Ø	Ø	PB *Nkákà “pangolin” (1684) PB *gòbò “skin” (1479)	PSB *ka PSB *kobo
	*k	h ~k	Ø ~k	Ø ~k	k	PB *gòngá “spear” (1448)	PSB *kɔngɔ
						PB *jògù “elephant” (1607)	PSB *yɔku
						PB *jíg “learn” (3338)	PSB *yok

> PSB *l occurred to fill the void created by the rule PB *d > *l > Ø in the same context (but we need to recall that PB *t > PSB *t in all contexts so we do not have an explanation to why some words with PB *t > PSB *l before non-high vowels). Moreover, we can note that some languages such as Mokpe or Bubia present PSB *l > Ø or /l/ which could also indicate that the process of deleting PSB *l is continuing.

The same phenomenon is observed with PB *k. When PB *k is deleted, Sawabantu languages (as well as the remaining NW Bantu languages A and some B20) reintroduced the missing sound with the sound change PB *g > PSB *k. Interestingly, PSB *k is undergoing lenition again in Sawabantu languages.

In conclusion, the examination of Table 26 highlights interesting sound shifts from PB to Sawabantu languages with strategies of replacement for sounds that have been deleted.

4 Conclusion

Sawabantu languages are fascinating for several reasons highlighted in this study. First, the study and reconstruction of their consonants evidences some interesting phenomena occurring at the PSB level with some PSB consonants undergoing lenition (PSB *p > /ɸ/, *k > Ø, *b > /β/ or /w/, *d > /l/). The study of PSB sound changes divided the Sawabantu languages into three subgroups: South, Central and Southwest-North. At the PB level, some interesting sound changes are observed with PB *t > PSB *t and *l, PB *k > Ø, PB *g > PSB *k and PB *d > Ø when followed by non-high vowels confirming that languages spoken in this particular region underwent many sound changes that resulted in the modification of the shape of their lexicon in comparison with PB (e.g., PB *kútà BLR3 2138 ‘oil’ > PSB *ula but [ma-kuta] in F31 Nilamba (data from Yukawa 1989) - a Bantu language spoken in Tanzania, PB *dók BLR3 1179 “vomit” > PSB *o but [kwi-lòka] in F31 Nilamba). Finally, this study confirms the different status of Manenguba languages (A15) that underwent different sound changes in C1 and C2 positions which led Nurse & Philippson (2003) to exclude these languages from the Sawabantu clade.

Acknowledgements

We wish to thank the two anonymous reviewers for their useful feedback. First author’s research was supported by the National Science Foundation (NSF), award 2152822.

References

- Atindogbe, Gratiën. 2013. A grammatical sketch of Mòkpè (Bakweri), Bantu A20. *African Study Monographs* 45. 1–163.
- Bastin, Yvonne, André Coupez & Michael Mann. 1999. *Continuity and divergence in the Bantu languages: Perspectives from a lexicostatistic study*, vol. 162. Tervuren: Musée royal de l'Afrique Centrale.
- Bastin, Yvonne, André Coupez, Kankawa E. Mumba & Thilo C. Schadeberg. 2002. *Bantu lexical reconstructions* 3. Tervuren: Africamuseum. https://www.africamuseum.be/de/research/discover/human_sciences/culture_society/blr.
- Bostoen, Koen. 2007. Pots, words and the Bantu problem: On lexical reconstruction and early African history. *Journal of African History* 48. 173–99.
- Bostoen, Koen. 2019. Reconstructing Proto-Bantu. In Mark Van de Velde, Koen Bostoen, Derek Nurse & Gérard Philippson (eds.), *The Bantu languages*, 2nd edn., 308–334. London: Routledge. DOI: 10.4324/9781315755946-10.
- Bostoen, Koen & Jean-Pierre Donzo. 2013. Bantu-Ubangi language contact and the origin of labial-velar stops in Lingombe (Bantu, C41, DRC). *Diachronica* 30. 435–68.
- Bybee, Joan & Shelece Easterday. 2019. Consonant strengthening: A crosslinguistic survey and articulatory proposal. *Linguistic Typology* 23. 263–302.
- Cahill, Michael. 1999. Aspects of the phonology of labial-velar stops. *Studies in African Linguistics* 28. 155–84.
- Campbell, Lyle. 1998. *Historical linguistics: An introduction*. Cambridge: MIT Press.
- Clements, George N. & Annie Rialland. 2008. Africa as a phonological area. In Bernard Heine & Derek Nurse (eds.), *A linguistic geography of Africa*, 36–85. Cambridge: Cambridge.
- Connell, Bruce. 1994. The structure of labial-velar stops. *Journal of Phonetics* 22(4). 441–476.
- Connell, Bruce. 1998. Feature geometry and the formation of labial-velars: A reply to Mutaka and Ebobissé. *Journal of West African Languages* 27. 17–32.
- Currie, Thomas E., Andrew Meade, Myrtille Guillon & Ruth Mace. 2013. Cultural phylogeography of the Bantu languages of sub-Saharan Africa. *Proceedings of the Royal Society B* 280.
- Eberhard, David M., Gary F. Simons & Charles D. Fennig. 2019. *Ethnologue: Languages of Asia*. <http://www.ethnologue.com/>.
- Ebobissé, Carl. 1989. Dialectométrie lexicale des parlers sawabantu. *Journal of West African Languages* 19. 57–66.

- Ebobissé, Carl. 2015. *Sawabantu: Eine Vergleichende Untersuchung Der Kuestensprachen Kameruns* (Schriften zur Afrikanistik 24). Frankfurt am Main: Peter Lang.
- Friesen, Dan T. 2002. *Oroko orthography development: Linguistic and sociolinguistic factors*. University of North Dakota. (MA thesis).
- Grollemund, Rebecca. 2012. *Nouvelles approches en classification: Application aux langues Bantu du Nord-Ouest*. Université Lumière Lyon 2. (Doctoral dissertation).
- Grollemund, Rebecca, Simon Branford, Koen Bostoen, Andrew Meade, Chris Venditti & Mark Pagel. 2015. Bantu expansion shows that habitat alters the route and pace of human dispersals. *PNAS* 112(43). 13296–13301.
- Grollemund, Rebecca & Gérard Philippson. to appear. A new definition of North-Western Bantu languages based on the study of Proto-Bantu *k and *g. In Rebecca Grollemund, Derek Nurse & John R. Watters (eds.), *Bantoid and Bantu in Cameroon: An historical re-assessment*. Berlin: De Gruyter Mouton. Forthcoming.
- Guthrie, Malcolm. 1967–1971. *Comparative Bantu: An introduction to the comparative linguistics and prehistory of the Bantu languages*. Gregg International Publishers Ltd.
- Hedinger, Robert. 1987. *The Manenguba languages*. London: School of Oriental & African Studies.
- Hyman, Larry M. 2019. More reflections on the nasal classes in Bantu. In John R. Watters (ed.), *Eastern Benue-Congo: Nouns, pronouns and verbs*, 225–38. Berlin: Language Science Press.
- Idiatov, Dmitry & Mark Van de Velde. 2021. The lexical distribution of labial-velar stops is a window into the linguistic prehistory of Northern Sub-Saharan Africa. *Language* 97(1). 72–107.
- Janssens, Baudoin. 1991. Doubles-réflexes apparents en ewondo ou les chassés croisés de la dérivation. *Pholia, Laboratoire de phonétique et linguistique africaine* 6. 155–80.
- Janssens, Baudoin. 1992. *Doubles reflexes consonantiques: Quatre études sur le Bantou de zone a: (Bubi, Nen, Bafia, Ewondo)*. Université Libre de Bruxelles. (Doctoral dissertation).
- Koile, Ezequiel, Simon J. Greenhill, Damián E. Blasi, Remco Bouckaert & Russell D. Gray. 2022. Phylogeographic analysis of the Bantu language expansion supports a rainforest route. *PNAS* 119(32).
- Leynseele, Helene Van & John M. Stewart. 1980. Harmonie consonantique en pré-nen. In Laurence Bouquiaux (ed.), *L'expansion bantoue: Actes du colloque international du CNRS*, vol. 2, 387–400. Viviers: Peeters.

- Maho, Jouni. N.d. *NUGL online: The online version of the new updated Guthrie list, a referential classification of the Bantu languages*.
- Meeussen, A. E. 1967. Bantu grammatical reconstructions. *Africana Linguistica* 3. 79–121.
- Möhlig, Wilhelm J. G. 1981. Stratification in the history of the Bantu languages. *Sprache und Geschichte in Afrika* 3. 251–317.
- Nurse, Derek & Gérard Philippson. 2003. Towards a historical classification of the Bantu languages. In Derek Nurse & Gérard Philippson (eds.), *The Bantu languages*, 164–181. London: Routledge.
- Pacchiarotti, Sara & Koen Bostoen. 2020. Proto-West-Coastal Bantu velar merger. *Africana Linguistica* 26. 139–195.
- Pacchiarotti, Sara & Koen Bostoen. 2022. Erratic velars in West-Coastal Bantu: Explaining irregular sound change in Central Africa. *Journal of Historical Linguistics* 12(3). 381–445.
- Philippson, Gérard. 2022. ‘Double reflexes’ in northwestern Bantu clarified. In Koen Bostoen, Gilles-Maurice de Schryver, Rozenne Guérois & Sara Pacchiarotti (eds.), *On reconstructing Proto-Bantu grammar*, 1–58. Berlin: Language Science Press.
- Stewart, John M. 1973. The lenis stops of the Potou Lagoon languages and their significance for pre-Bantu reconstruction. In Mary Elizabeth Kropp Dakubu (ed.), *Papers in Ghanian linguistics*, vol. 49 (Transactions of the Linguistic Circle of Accra II). Legon: Institute of African Studies, University of Ghana.
- Stewart, John M. 1975. Lenis stops and the origins of Volta-Comoe consonant mutation. In R. K. Herbert (ed.), *Proceedings of the Sixth Conference on African Linguistics*, 16–31. Columbus: Ohio State University.
- Stewart, John M. 1993. The second Tano consonant shift and its likeness to Grimm’s Law. *Journal of West African Languages* 23. 3–39.
- Wills, Jeffrey. 2022. Sorting out Proto-Bantu *j. In Koen Bostoen, Gilles-Maurice de Schryver, Rozenne Guérois & Sara Pacchiarotti (eds.), *On reconstructing Proto-Bantu grammar*, 59–101. Berlin: Language Science Press.
- Yukawa, Yasutoshi. 1989. *A classified vocabulary of the Nilamba language* (Bantu Vocabulary Series 5). Tokyo: Institute for the Study of Languages, Cultures of Asia & Africa.